

## MIXTURES OF SOME DISCRETE DISTRIBUTIONS

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### 1. Introduction

If  $F = \{F\}$  is a family of distributions and  $\mu$  is a measure on a Borel field of subsets of  $F$  with  $\mu(F) = 1$ , then  $H(\cdot) = \int F(\cdot) d\mu(F)$  is again a distribution function called a  $\mu$ -mixture of  $F$ . In case of absolutely continuous distributions, the mixture can be written as  $\int f(x, \theta) g(\theta) d\theta$ , where  $g(\theta)$  is the density of the parameter  $\theta$  which indexes the family  $\{f\}$  of probability density functions.

In this paper we have obtained mixtures of multivariate binomial, bivariate multinomial and negative multinomial with generalized beta distribution introduced by Holla (1968).

Skellam (1948) and later Ishi and Hayakawa (1961) obtained a mixture of binomial distribution treating  $p$  the probability of success, as a random variable with beta distribution. Holla (1968) replaced the beta distribution by generalized beta. Ishi and Hayakawa (1961) also obtained a mixture of bivariate binomial and beta distribution. In section 3, we obtain a mixture of multivariate binomial and generalized beta and give the factorial moments of the mixture. Some of the conditional distributions and the distribution of the sum of the components is discussed. Ishi and Hayakawa (1961) obtained mixtures of multinomial distributions and beta distribution. This result has been generalized in section 4 with beta replaced by generalized Dirichlet; factorial moments of the mixture and conditional distribution are discussed.

Holla (1968) has obtained a mixture of negative binomial and generalized beta. In section 5, we obtain a mixture of negative multinomial, discussed by Sibuya, Yoshimura and Shimizu (1969), and generalized Dirichlet distribution; its factorial moments are then discussed. It may be remarked that in all these cases, characteristic function is not very convenient to obtain the moments.

### 2. Some Standard Results

1) A r.v.  $X$  is said to have a generalized beta distribution if its p.d.f. is of the form

$$(2.1) \quad f(x; \beta, \nu_1, \nu_2) = \frac{1}{B(\nu_1, \nu_2)} \frac{\exp(-\beta x) x^{\nu_1-1} (1-x)^{\nu_2-1}}{\Phi(\nu_1: \nu_1 + \nu_2; -\beta)}, \quad 0 < x < 1, \nu_1, \nu_2 > 0, \beta \geq 0;$$

where  $\Phi$  is the confluent hypergeometric function. The fact that this is a p.d.f. can be verified by using

$$(2.2) \quad \Phi(a; b; z) = \frac{\Gamma(b)}{\Gamma(a)\Gamma(b-a)} \int_0^1 \exp(tz) t^{a-1} (1-t)^{b-a-1} dt,$$

$\text{Re } b > \text{Re } a > 0$ ; Rainville (1960). A straightforward generalization to the multivariate case is

- 2) A r.v.  $(X_1, X_2, \dots, X_k)$  is said to have multivariate generalized beta (generalized Dirichlet) distribution if its p.d.f. is of the form

$$(2.3) \quad f(x_1, x_2, \dots, x_k) = \frac{1}{B(\nu_1, \dots, \nu_{k+1})} \cdot \frac{\exp(-\beta \sum_{i=1}^k x_i) x_1^{\nu_1-1} \dots x_k^{\nu_k-1} (1-x_1-\dots-x_k)^{\nu_{k+1}-1}}{\Phi(\sum_{i=1}^k \nu_i; \sum_{i=1}^{k+1} \nu_i; -\beta)},$$

with

$$x_i > 0, \sum_{i=1}^k x_i < 1, \nu_i > 0; i = 1, 2, \dots, k;$$

where

$$(2.4) \quad B(\nu_1, \dots, \nu_{k+1}) = \frac{\Gamma(\nu_1) \Gamma(\nu_2) \dots \Gamma(\nu_{k+1})}{\Gamma(\nu_1 + \nu_2 + \dots + \nu_{k+1})}.$$

### 3. Multivariate Binomial Generalized Beta Distributions

$k$  variate binomial-generalized beta distribution (k-B.G.B.D.) is given by

$$(3.1) \quad h(x_1, x_2, \dots, x_k) = \int_0^1 \binom{n_1}{x_1} \binom{n_2}{x_2} \dots \binom{n_k}{x_k} p^{\sum_{i=1}^k x_i} (1-p)^{n_1 - \sum_{i=1}^k x_i} g(p) dp$$

where  $g(p)$  is the p.d.f. of a r.v. with generalized beta distribution.

Thus

$$(3.2) \quad h(x_1, x_2, \dots, x_k) = \frac{\binom{n_1}{x_1} \binom{n_2}{x_2} \dots \binom{n_k}{x_k}}{B(\nu_1, \nu_2) \Phi(\nu_1; \nu_1 + \nu_2; -\beta)} \int_0^1 \exp(-\beta p) p^{\sum_{i=1}^k x_i + \nu_1 - 1} (1-p)^{\sum_{i=1}^k n_i - \sum_{i=1}^k x_i + \nu_2 - 1} dp$$

$$= \binom{n_1}{x_1} \binom{n_2}{x_2} \dots \binom{n_k}{x_k} \frac{B(\sum_{i=1}^k x_i + \nu_1, \sum_{i=1}^k n_i - \sum_{i=1}^k x_i + \nu_2)}{B(\nu_1, \nu_2)} \cdot \frac{\Phi(\sum_{i=1}^k x_i + \nu_1; \sum_{i=1}^k n_i + \nu_1 + \nu_2; -\beta)}{\Phi(\nu_1; \nu_1 + \nu_2; -\beta)}$$

$$x_i = 0, 1, 2, \dots, n_i; i = 1, 2, \dots, k.$$

In particular, the bivariate binomial-generalized beta distribution is given by

$$(3.3) \quad h(x_1, x_2) = \frac{B(x_1 + x_2 + \nu_1, n_1 + n_2 - x_1 - x_2 + \nu_2)}{(n_1 + 1)(n_2 + 1) B(\nu_1, \nu_2) B(x_1 + 1, n_1 - x_1 + 1) B(x_2 + 1, n_2 - x_2 + 1)} \cdot \frac{\Phi(x_1 + x_2 + \nu_1; n_1 + n_2 + \nu_1 + \nu_2; -\beta)}{\Phi(\nu_1; \nu_1 + \nu_2; -\beta)}$$

For  $\beta = 0$ , this conforms with (6) of Ishi and Hayakawa (1961).

To find the factorial moment of the k-B.G.B.D., we notice that  $r_1 \dots r_k$ -th factorial moment of k variate binomial distribution is

$$\begin{matrix} [r_1] & [r_k] & \sum_{i=1}^k r_i \\ n_1 & \dots n_k & \cdot p \end{matrix}$$

Thus the  $r_1 \dots r_k$ -th factorial moment of k-B.G.B.D. is

$$(3.4) \quad E(X_1^{[r_1]} \dots X_k^{[r_k]}) = \int_0^1 \begin{matrix} [r_1] & [r_k] & \sum_{i=1}^k r_i \\ n_1 & \dots n_k & p \end{matrix} g(p) dp = \begin{matrix} [r_1] & [r_k] \\ n_1 & \dots n_k \end{matrix} \frac{B(\sum_{i=1}^k r_i + \nu_1, \nu_2) \Phi(\sum_{i=1}^k r_i + \nu_1; \sum_{i=1}^k r_i + \nu_1 + \nu_2; -\beta)}{B(\nu_1, \nu_2) \Phi(\nu_1; \nu_1 + \nu_2; -\beta)}$$

and in particular

$$(3.5) \quad E(X_1, X_2, \dots, X_k) = n_1 \cdot n_2 \dots n_k \frac{(\nu_1 + k - 1)^{[k]} \Phi(\nu_1 + k; \nu_1 + \nu_2 + k; -\beta)}{(\nu_1 + \nu_2 + k - 1)^{[k]} \Phi(\nu_1; \nu_1 + \nu_2; -\beta)}$$

The conditional distribution of  $X_2, X_3, \dots, X_k$  given  $X_1$  is obtained as

$$h(x_2, \dots, x_k | x_1) = \frac{h(x_1, x_2, \dots, x_k)}{h(x_1)}, \text{ where}$$

$$h(x_1) = \binom{n_1}{x_1} \frac{B(x_1 + \nu_1, n_1 + \nu_2 - x_1)}{\beta(\nu_1, \nu_2)} \cdot \frac{\Phi(x_1 + \nu_1; n_1 + \nu_1 + \nu_2; -\beta)}{\Phi(\nu_1, \nu_1 + \nu_2; -\beta)}$$

Thus

$$(3.6) \quad h(x_2, \dots, x_k | x_1) = \binom{n_2}{x_2} \dots \binom{n_k}{x_k} \frac{B(\sum_{i=1}^k x_i + \nu_1, \sum_{i=1}^k n_i - \sum_{i=1}^k x_i + \nu_2)}{B(x_1 + \nu_1, n_1 + \nu_2 - x_1)} \\ \cdot \frac{\Phi(\sum_{i=1}^k x_i + \nu_1; \sum_{i=1}^k n_i + \nu_1 + \nu_2; -\beta)}{\Phi(x_1 + \nu_1; n_1 + \nu_1 + \nu_2; -\beta)}$$

Thus the conditional distribution of  $X_2, \dots, X_k$  given  $X_1$  is  $(k-1)$  variate binomial generalized beta, with mean

$$(3.7) \quad E(X_2, \dots, X_k | X_1) = n_2 n_3 \dots n_k \frac{(\nu_1 + x_1 + k - 2)^{[k-1]}}{(\nu_1 + \nu_2 + n_1 + k - 2)^{[k-1]}} \\ \cdot \frac{\Phi(\nu_1 + x_1 + k - 1; \nu_1 + \nu_2 + n_1 + k - 1; -\beta)}{\Phi(\nu_1 + x_1; \nu_1 + \nu_2 + n_1; -\beta)}$$

The conditional distribution of  $X_k$  given  $X_1, \dots, X_{k-1}$  is obtained as

$$(3.8) \quad h(x_k | x_1, \dots, x_{k-1}) = \frac{h(x_1, x_2, \dots, x_k)}{h(x_1, x_2, \dots, x_{k-1})} \\ = \binom{n_k}{x_k} \frac{B(\sum_{i=1}^k x_i + \nu_1, \sum_{i=1}^k n_i - \sum_{i=1}^k x_i + \nu_2)}{B(\sum_{i=1}^{k-1} n_i + \nu_1, \sum_{i=1}^{k-1} n_i - \sum_{i=1}^{k-1} x_i + \nu_2)} \\ \cdot \frac{\Phi(\sum_{i=1}^k x_i + \nu_1; \sum_{i=1}^k n_i + \nu_1 + \nu_2; -\beta)}{\Phi(\sum_{i=1}^{k-1} x_i + \nu_1; \sum_{i=1}^{k-1} n_i + \nu_1 + \nu_2; -\beta)}$$

which is binomial generalized beta distribution. Regression of  $X_k$  on  $X_i$  ( $i=1,2,\dots,k-1$ ) is

$$(3.9) E(X_k | X_1, X_2, \dots, X_{k-1})$$

$$= \frac{n_k \binom{k-1}{\sum_{i=1}^{k-1} x_i + \nu_1} \Phi\left(\sum_{i=1}^{k-1} x_i + \nu_1 + 1; \sum_{i=1}^{k-1} n_i + \nu_1 + \nu_2 + 1; -\beta\right)}{\sum_{i=1}^{k-1} n_i + \nu_1 + \nu_2 \Phi\left(\sum_{i=1}^{k-1} x_i + \nu_1; \sum_{i=1}^{k-1} n_i + \nu_1 + \nu_2; -\beta\right)}$$

The distribution of  $Z = \sum_{i=1}^k X_i$  is given by

$$(3.10) h(z) = \sum_{x_1, x_2, \dots, x_{k-1}} \binom{n_1}{x_1} \binom{n_2}{x_2} \dots \binom{n_{k-1}}{x_{k-1}} \binom{n_k}{z - x_1 - x_2 - \dots - x_{k-1}}$$

$$= \frac{B(z + \nu_1, \sum_{i=1}^k n_i - z + \nu_2) \Phi(z + \nu_1; \sum_{i=1}^k n_i + \nu_1 + \nu_2; -\beta)}{B(\nu_1, \nu_2) \Phi(\nu_1; \nu_1 + \nu_2; -\beta)}$$

$$= \frac{B(z + \nu_1, \sum_{i=1}^k n_i - z + \nu_2) \Phi(z + \nu_1; \sum_{i=1}^k n_i + \nu_1 + \nu_2; -\beta)}{\left(\sum_{i=1}^k n_i + 1\right) B(\nu_1, \nu_2) B(z + 1, \sum_{i=1}^k n_i - z + 1) \Phi(\nu_1; \nu_1 + \nu_2; -\beta)}$$

which is B-G.B.D.

#### Remark

For  $\beta = 0$ , the above result gives us the results of Ishi and Hayakawa (1961) when  $k=2$  and  $k=3$ .

#### 4. Bivariate multinomial-generalized Dirichlet distribution

The bivariate multinomial-generalized Dirichlet distribution (B.M-G.D.D.) is given by

$$(4.1) h(x, y) = \int \frac{n!}{x! y! (n-x-y)!} p^x q^y (1-p-q)^{n-x-y} g(p, q) dp dq$$

$$x = 0, 1, 2, \dots, n$$

$$y = 0, 1, 2, \dots, n, x+y \leq n$$

$$0 \leq p < 1, 0 \leq q < 1$$

where  $g(p, q)$  is the p.d.f. of a r.v. with generalized Dirichlet distribution. Thus

$$\begin{aligned}
 (4.2) \quad h(x, y) &= \frac{n!}{x! y! (n-x-y)!} \cdot \frac{\Gamma(\nu_1 + \nu_2 + \nu_3)}{\Gamma(\nu_1)\Gamma(\nu_2)\Gamma(\nu_3)} \cdot \frac{1}{\Phi(\nu_1 + \nu_2; \nu_1 + \nu_2 + \nu_3; -\beta)} \\
 &\quad \cdot \int \int \exp(-\beta(p+q)) p^{x+\nu_1-1} q^{y+\nu_2-1} (1-p-q)^{n+\nu_3-x-y-1} dp dq \\
 &= \frac{B(x+\nu_1, y+\nu_2, n+\nu_3-x-y)}{(n+2)(n+1)B(1+x, 1+y, n+1-x-y)B(\nu_1, \nu_2, \nu_3)} \cdot \\
 &\quad \cdot \frac{\Phi(x+y+\nu_1+\nu_2; n+\nu_1+\nu_2+\nu_3; -\beta)}{\Phi(\nu_1+\nu_2; \nu_1+\nu_2+\nu_3; -\beta)}
 \end{aligned}$$

For  $\beta = 0$ , this gives (17) of Ishi and Hayakawa (1961).

To find the factorial moment of this distribution, we notice that  $r$ -th factorial moment of the bivariate multinomial distribution is  $n^{[r+s]} p^r q^s$ . Thus  $r$ -th factorial moment of B.M-G.D.D. is

$$\begin{aligned}
 (4.3) \quad E(X^{[r]} Y^{[s]}) &= \int \int n^{[r+s]} p^r q^s g(p, q) dp dq \\
 &= \frac{n^{[r+s]}}{B(\nu_1, \nu_2, \nu_3) \Phi(\nu_1 + \nu_2; \nu_1 + \nu_2 + \nu_3; -\beta)} \cdot \\
 &\quad \cdot \int \int \exp(-\beta(p+q)) p^{r+\nu_1-1} q^{s+\nu_2-1} (1-p-q)^{\nu_3-1} dp dq \\
 &= n^{[r+s]} \frac{B(r+\nu_1, s+\nu_2, \nu_3)}{B(\nu_1, \nu_2, \nu_3)} \cdot \frac{\Phi(r+s+\nu_1+\nu_2; r+s+\nu_1+\nu_2+\nu_3; -\beta)}{\Phi(\nu_1+\nu_2; \nu_1+\nu_2+\nu_3; -\beta)}
 \end{aligned}$$

For  $\beta = 0$ , this gives corresponding result of Ishi and Hayakawa (1961). In particular

$$(4.4) \quad E(X Y) = n(n-1) \frac{\nu_1 \nu_2}{(\nu_1 + \nu_2 + \nu_3 + 1)(\nu_1 + \nu_2 + \nu_3)} \cdot \frac{\Phi(2+\nu_1+\nu_2; 2+\nu_1+\nu_2+\nu_3; -\beta)}{\Phi(\nu_1+\nu_2; \nu_1+\nu_2+\nu_3; -\beta)}$$

$$(4.5) E(X) = \frac{n \nu_1}{\nu_1 + \nu_2 + \nu_3} \frac{\Phi(1 + \nu_1 + \nu_2; 1 + \nu_1 + \nu_2 + \nu_3; -\beta)}{\Phi(\nu_1 + \nu_2; \nu_1 + \nu_2 + \nu_3; -\beta)}$$

and

$$(4.6) E(Y) = \frac{n \nu_2}{\nu_1 + \nu_2 + \nu_3} \frac{\Phi(1 + \nu_1 + \nu_2; 1 + \nu_1 + \nu_2 + \nu_3; -\beta)}{\Phi(\nu_1 + \nu_2; \nu_1 + \nu_2 + \nu_3; -\beta)}$$

Also we observe the following identity

$$(4.7) \sum_{y=0}^n \sum_{x=0}^{n-y} x[r]_y[s] \frac{B(x + \nu_1, y + \nu_2, n + \nu_3 - x - y)}{(n+2)(n+1)B(1+x, 1+y, n+1-x-y)} \\ \cdot \Phi(x+y+\nu_1+\nu_2; n+\nu_1+\nu_2+\nu_3; -\beta) \\ = n[r+s] B(r+\nu_1, s+\nu_2, \nu_3) \Phi(r+s+\nu_1+\nu_2; r+s+\nu_1+\nu_2+\nu_3; -\beta)$$

The marginal distribution of X is

$$(4.8) h(x) = \frac{1}{B(\nu_1, \nu_2)(n+1)} \frac{B(x+\nu_1, n+\nu_2-x)}{B(1+x, n+1-x)} \frac{\Phi(x+\nu_1; n+\nu_1+\nu_2; -\beta)}{\Phi(\nu_1; \nu_1+\nu_2; -\beta)},$$

$$x = 0, 1, 2, \dots, n; \nu_1, \nu_2 > 0; \beta \geq 0$$

which is B-G.B.D. See equation (16) of Holla (1968). The conditional distribution of Y given X is given by

$$(4.9) h(y|x) = \frac{h(x,y)}{h(x)} \\ = \frac{B(n+\nu_1+\nu_2, \nu_3)}{(n+1-x)B(\nu_1+\nu_2, \nu_3)B(n+\nu_2-x, \nu_3)} \frac{B(\nu_2+y, n+\nu_3-x-y)}{B(1+y, n+1-x-y)} \\ \cdot \frac{\Phi(x+y+\nu_1+\nu_2; n+\nu_1+\nu_2+\nu_3; -\beta)}{\Phi(\nu_1+\nu_2; \nu_1+\nu_2+\nu_3; -\beta)} \frac{\Phi(\nu_1; \nu_1+\nu_2; -\beta)}{\Phi(x+\nu_1; n+\nu_1+\nu_2; -\beta)}$$

For  $\beta = 0$  this gives corresponding result of Ishi and Hayakawa (1961), which seems to be in error.

The distribution of  $Z = X + Y$  is given by

$$(4.10) \quad g(z) = \frac{1}{(n+1)B(\nu_1+\nu_2, \nu_3)} \frac{B(\nu_1+\nu_2+z, n+\nu_3-z)}{B(1+z, n+1-z)} \frac{\Phi(z+\nu_1+\nu_2; n+\nu_1+\nu_2+\nu_3; -\beta)}{\Phi(\nu_1+\nu_2; \nu_1+\nu_2+\nu_3; -\beta)}$$

This is B-G.B.D. and gives for  $\beta = 0$ , corresponding result of Ishi and Hayakawa (1961).

### 5. Negative multinomial generalized Dirichlet distribution

The negative multinomial generalized Dirichlet distribution (N.M-G.D.D.) is given by

$$(5.1) \quad h(x_1, x_2, \dots, x_r) = \int f(x_1, \dots, x_r; p_1, \dots, p_r) g(p_1, \dots, p_r) dp_1 \dots dp_r$$

$$x_i = 0, 1, 2, \dots; i = 1, 2, \dots, r;$$

where

$$(5.2) \quad f(x_1, \dots, x_r) = \frac{\Gamma(k+x_1+x_2+\dots+x_r)}{\Gamma(k)x_1! x_2! \dots x_r!} p_1^{x_1} \dots p_r^{x_r} (1-p_1-\dots-p_r)^k$$

$$x_i = 0, 1, 2, \dots; i = 1, 2, \dots, r$$

and

$$(5.3) \quad g(p_1, \dots, p_r) = \frac{\Gamma(\nu_1 + \dots + \nu_{r+1})}{\Gamma(\nu_1)\Gamma(\nu_2)\dots\Gamma(\nu_{r+1})}$$

$$\frac{\exp(-\beta(\sum_{i=1}^r p_i)) p_1^{\nu_1-1} \dots p_r^{\nu_r-1} (1-p_1-\dots-p_r)^{\nu_{r+1}-1}}{\Phi(\sum_{i=1}^r \nu_i; \sum_{i=1}^{r+1} \nu_i; -\beta)}$$

$$p_i > 0, \sum_{i=1}^r p_i < 1, \nu_i > 0, i = 1, 2, \dots, r+1.$$

Thus we obtain

$$(5.4) \quad h(x_1, x_2, \dots, x_r) = \frac{\Gamma(k+x_1+\dots+x_r)}{\Gamma(k)x_1! x_2! \dots x_r!} \frac{B(x_1+\nu_1, \dots, x_r+\nu_r, k+\nu_{r+1})}{B(\nu_1, \nu_2, \dots, \nu_{r+1})}$$

$$\cdot \frac{\Phi(\sum_{i=1}^r (x_i+\nu_i); \sum_{i=1}^{r+1} \nu_i + \sum_{i=1}^r x_i + k; -\beta)}{\Phi(\sum_{i=1}^r \nu_i; \sum_{i=1}^{r+1} \nu_i; -\beta)}$$

The  $j_1, \dots, j_r$ -th factorial moment of N.M.-G.D.D. is given by

$$(5.5) \quad E(X_1^{[j_1]} \dots X_r^{[j_r]})$$

$$= \int \frac{p_1^{j_1} \dots p_r^{j_r}}{(1-p_1-p_2-\dots-p_r)^{j_1+j_2+\dots+j_r}} (j_1+j_2+\dots+j_r+k-1)^{[j_1+\dots+j_r]} \cdot g(p_1, \dots, p_r) dp_1 \dots dp_r$$

$$= \frac{(j_1+j_2+\dots+j_r+k-1)^{[j_1+\dots+j_r]}}{B(\nu_1, \dots, \nu_{r+1}) (\sum_{i=1}^r \nu_i; \sum_{i=1}^{r+1} \nu_i; -\beta)}$$

$$\cdot \int \exp(-\beta (\sum_{i=1}^r p_i) p_1^{j_1+\nu_1-1} \dots p_r^{j_r+\nu_r-1} (1-p_1-\dots-p_r)^{\nu_{r+1}-(j_1+\dots+j_r)})$$

$$dp_1 \dots dp_r$$

$$= (j_1+j_2+\dots+j_r+k-1)^{[j_1+\dots+j_r]} \frac{B(j_1+\nu_1, \dots, j_r+\nu_r, \nu_{r+1} - \sum_{i=1}^r j_i)}{B(\nu_1, \nu_2, \dots, \nu_{r+1})}$$

$$\frac{\Phi(\sum_{i=1}^r (j_i+\nu_i); \sum_{i=1}^{r+1} \nu_i; -\beta)}{\Phi(\sum_{i=1}^r \nu_i; \sum_{i=1}^{r+1} \nu_i; -\beta)}$$

In particular the mean of N.M-G.D.D. is

$$E(X_1, \dots, X_r) = (r+k-1) \frac{[r] \nu_1 \nu_2 \dots \nu_r \Gamma(\nu_{r+1}-r)}{\Gamma(\nu_{r+1})} \cdot \frac{\Phi(\sum_{i=1}^r \nu_i + r; \sum_{i=1}^{r+1} \nu_i; -\beta)}{\Phi(\sum_{i=1}^r \nu_i; \sum_{i=1}^{r+1} \nu_i; -\beta)}.$$

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